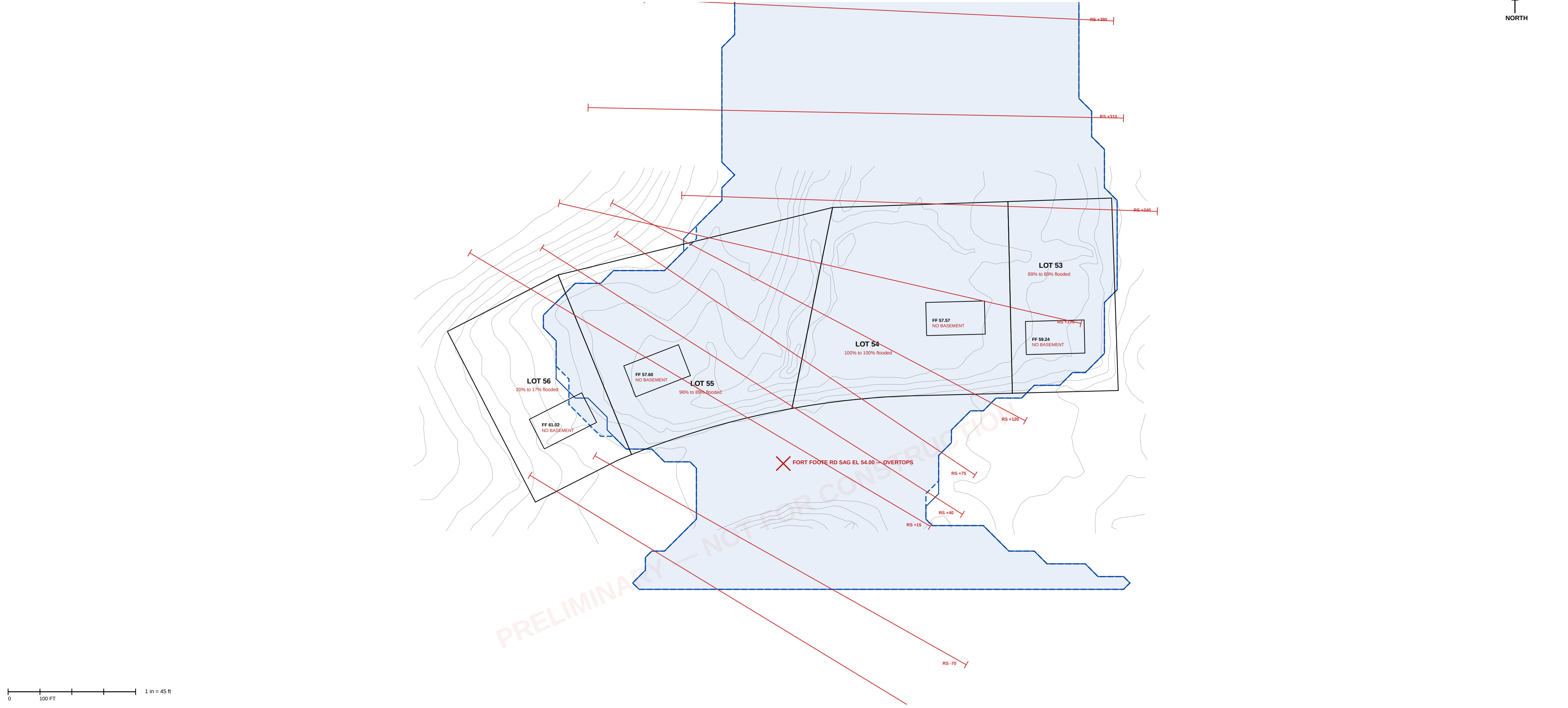
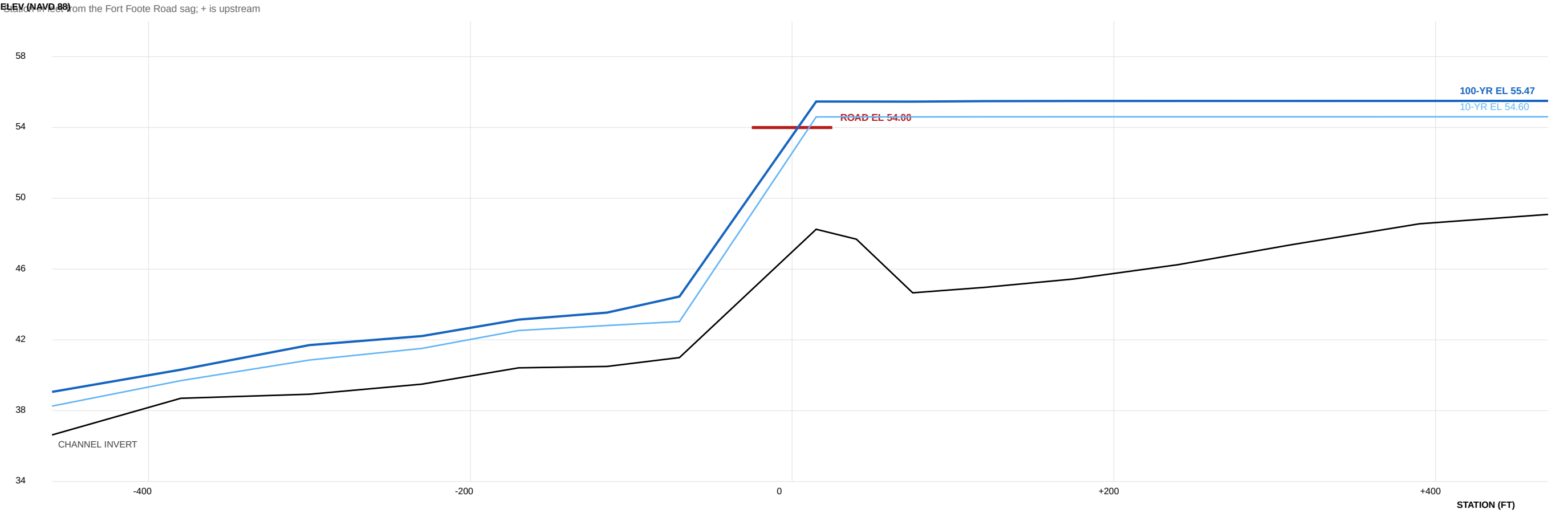


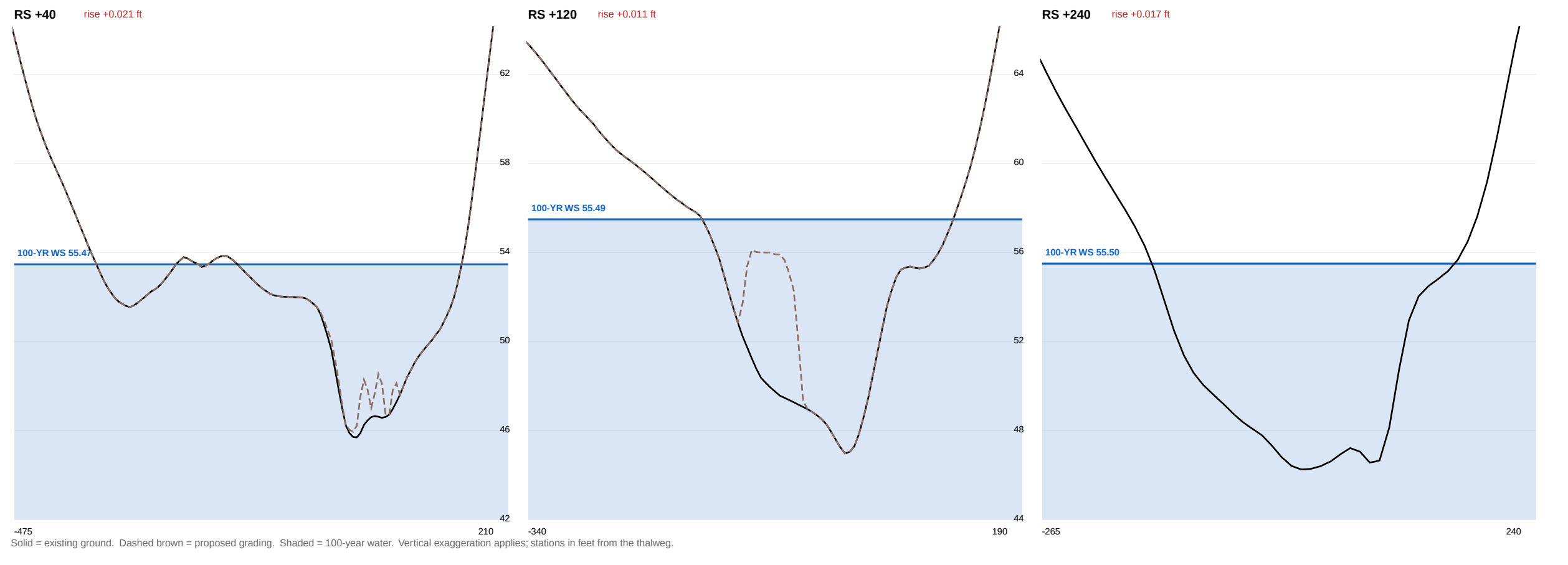
FLOODPLAIN DELINEATION — EXISTING AND PROPOSED, 100-YEAR



WATER-SURFACE PROFILE ALONG NORTH BRANCH BROAD CREEK



MODELLED CROSS SECTIONS — EXISTING AND PROPOSED



DESIGN DISCHARGES — NRCSTR-55, TYPE II					
STORM	P24 (IN)	Q (IN)	qw	PEAK (CFS)	
10-year	4.81	1.75	324	351	
25-year	6.01	2.63	350	571	
50-year	7.08	3.47	367	792	
100-year	8.29	4.48	382	1,061	
Area	397 ac	CN	68.0	Tc 0.86 hr	
CULVERT AT Q100					
36 in		HW 55.45	HWWD 3.6	939 over rd	
48 in	inlet	HW 55.36	HWWD 2.7	855 over rd	
60 in	inlet	HW 55.26	HWWD 2.1	757 over rd	
72 in	inlet	HW 55.13	HWWD 1.7	650 over rd	
84 in	inlet	HW 55.00	HWWD 1.5	541 over rd	
96 in	inlet	HW 54.87	HWWD 1.3	439 over rd	

EXISTING vs PROPOSED WATER SURFACE — 100-YEAR				
SECTION	EXISTING	PROPOSED	RISE (FT)	
RS +15	55.47	55.47	+0.000	
RS +40	55.47	55.49	+0.021	RISE
RS +175	55.46	55.49	+0.030	RISE
RS +120	55.49	55.50	+0.011	RISE
RS +175	55.50	55.51	+0.016	RISE
RS +240	55.50	55.52	+0.017	RISE
RS +310	55.50	55.52	+0.018	RISE
RS +390	55.50	55.52	+0.017	RISE
RS +470	55.51	55.52	+0.017	RISE
MAXIMUM			+0.030	NOT NO-RISE
Storage removed below the 100-yr surface			2,108 cy	
Total fill in the modelled reach			2,243 cy	
Compensatory storage			REQUIRED	

FLOODPLAIN ON THE LOTS, AND FREEBOARD — 100-YEAR				
LOT	AREA (SF)	FLOODED EXIST	FLOODED PROP	FILL < FLOOD
LOT 53	12,500	11,100 (89%)	11,100 (89%)	18 cy
LOT 54	23,500	23,500 (100%)	23,500 (100%)	1,142 cy
LOT 55	25,600	24,500 (96%)	22,100 (86%)	922 cy
LOT 56	13,900	2,800 (20%)	2,400 (17%)	12 cy
LOWEST FLOOR CHECK AGAINST WS EL 56.52				
LOT	FIN FLOOR	FREEBOARD	BASEMENT	FOUNDATION
LOT 53	59.24	+3.72	none	slab
LOT 54	57.57	+2.55	none	slab
LOT 55	57.60	+2.08	none	slab
LOT 56	61.02	+5.50	none	slab

BASIS, METHOD AND LIMITATIONS

HYDROLOGY NRCSTR-55 graphical peak discharge, Type II distribution. Drainage area delineated by D8 routing on the USGS 3DEP bare-earth DEM. Curve number computed from the county 2023 impervious surface and tree canopy layers and the NRCS soil survey. Rainfall from NOAA Atlas 14 Vol. 2 Ver. 3, partial duration series, at 38.7611 N 77.0115 W.

HYDRAULICS One-dimensional steady gradually-varied flow, standard step, conveyance subdivided at the bank stations, average conveyance friction slope. Crossing analyzed to FHWA HDS-5 with both controls computed and the roadway treated as a broad-crested weir.

TERRAIN POKas Contour 2 F (2023), NAVD 88, interpolated to a 10 ft grid. THIS IS NOT A FIELD SURVEY.

DATUM NAVD 88 + NGVD 29 - 0.86 ft here, from NGS marks HV4762, HV4763 and HV4728. On that basis the 57 ft WSSC of FPS-770017 is about EL 56.3 NAVD 88, against EL 55.47 computed. To be confirmed by leveling to benchmarks H31A, H32A and H32B on the field topographic survey.

NOT DONE No field survey. The existing culvert has never been measured and its size, inverts and entrance are assumed. HEC-RAS was not run and no HEC-RAS files exist. FPS 200546, the controlling study of record, has not been obtained. Upstream stormwater facilities were not inventoried. No floodway was delineated and none exists on current mapping. No unsteady or storage routing.

FLOODPLAIN OF RECORD FEMA FIRM panel 2403C02202E eff. 2016-09-16 maps these lots Zone X, outside the special flood hazard area, with no BFE and no floodway. That is a statement about FEMA mapping, not about flood risk. The county regulates this floodplain through the FPS series regardless.

EASEMENT The only patented floodplain easement in Indian Queen East (Pkt 118-083, recorded 17 January 1984, 5.32 ac) lies entirely south of Fort Foote Road and covers none of Lots 53-56.

KEALEE

Design, build, deliver

PROJECT
Indian Queen East — Lots 53, 54, 55 and 56

ADDRESS
9588 · 9584 · 9580 · 9576 Fort Foote Road

JURISDICTION
Prince George's County, Maryland

ZONE
RSF-95

PLAT OF RECORD
Plat Book WWW 65, folio 60

SHEET
FP-101 — FLOODPLAIN STUDY AND DELINEATION, EXISTING AND PROPOSED

DISCIPLINE
Maryland Professional Engineer

HORIZONTAL DATUM
NAD 83, Maryland State Plane, US survey feet (EPSG:2248)

VERTICAL DATUM
NAVD 88

DATE
2026-09-11

STATUS
PRELIMINARY FEASIBILITY. NOT A SEALED FLOODPLAIN STUDY. No field survey. Culvert never measured. HEC-RAS not run. FPS 200546, the controlling study of record, not obtained. No professional engineer has reviewed or sealed this sheet.

PROFESSIONAL CERTIFICATION

Maryland P.E. No. _____ Date _____

KEY FINDINGS

Contributing drainage area 397 ac (0.620 sq mi)

Composite curve number 68.0

Time of concentration 0.86 hr

Q100 at the crossing 1,061 cfs

Fort Foote Road sag EL 54.00

100-yr water surface, existing EL 55.47

100-yr water surface, proposed EL 55.52

Max rise, proposed vs existing +0.03 ft — NOT no-rise

Flood storage removed by fill 2,108 cy

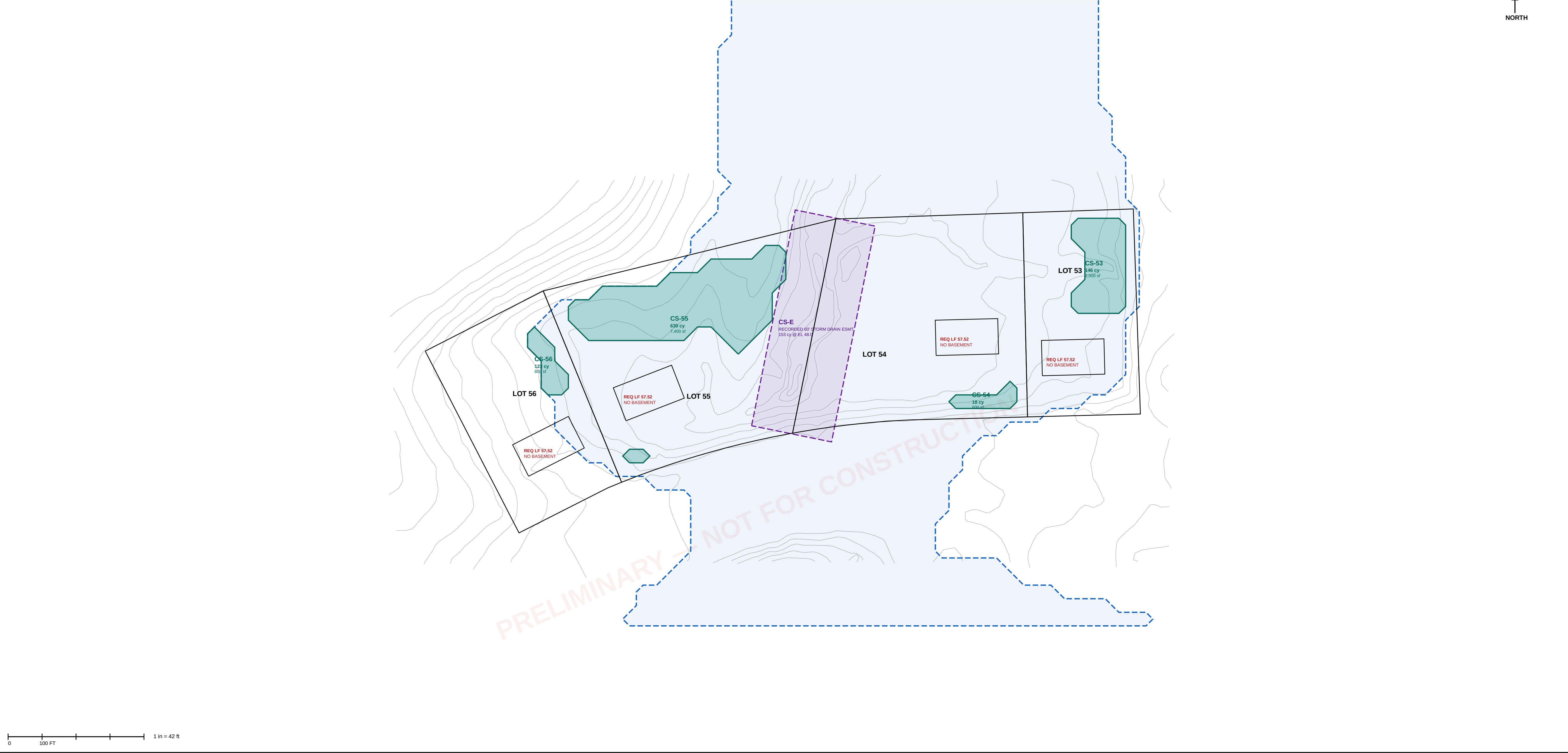
Road overtops at Q100 956 cfs over the pavement

LEGEND

- 100-yr floodplain limit, EXISTING
- 100-yr floodplain limit, PROPOSED
- Modelled cross section
- Lot boundary of record
- Existing contour (2 ft, county LIDAR)

COMPENSATORY STORAGE DESIGN — PLAN

Cells CS-53 to CS-56 excavated to EL 50.50; cell CS-E within the recorded 60 ft storm drain easement



KEALEE

Design, build, deliver

PROJECT

Indian Queen East — Lots 53, 54, 55 and 56

ADDRESS

9588 · 9584 · 9580 · 9576 Fort Foote Road

JURISDICTION

Prince George's County, Maryland

SHEET

FP-102 — MITIGATION ANALYSIS AND COMPENSATORY STORAGE DESIGN

DISCIPLINE

Maryland Professional Engineer

VERTICAL DATUM

NAVD 88

DATE

2026-09-11

STATUS

PRELIMINARY FEASIBILITY. NOT A SEALED FLOODPLAIN STUDY. Quantities are for concept evaluation and are not construction quantities. No professional engineer has reviewed or sealed this sheet.

PROFESSIONAL CERTIFICATION

Maryland P.E. No. _____ Date _____

DETERMINATIONS

RECOMMENDED: develop 53 and 56

Removal of all lots from floodplain

Controlling talwater, lower bound

Lowest ground, Lot 55

Basements

Required lowest floor

Compensation required

Compensation available on site

Crossing enlargement

Lot 54 flooded, 2-year event

Road overtopped

Storage to dry the rear yards

Upstream diversion benefit

53 + 56 only; compensation needed

available on 54 + 55 at EL 50

DEDICATE 54 AND 55

NOT ACHIEVABLE

EL 44.45

EL 44.66

EL 57.52

2,108 cy

1,070 cy

NOT RECOMMENDED

89%

BETWEEN 5- AND 10-YEAR

136,570 cy — NOT FEASIBLE

0.07 ft

31 cy

1,399 cy

DESIGN CRITERIA — COMPENSATORY STORAGE

1. Excavation floor EL 50.50, above the invert of the receiving channel, so each cell drains by gravity.

2. Cells located within the existing condition 100-year floodplain limit, being the area to be placed under floodplain easement.

3. Side slopes no steeper than 3:1, stabilized and revegetated, no retaining structure.

4. Minimum 10 ft from every property line and 20 ft from every structure.

5. Positive hydraulic connection to the recorded storm drain easement corridor at the cell invert.

6. Volume provided at the same elevation increments from which it is taken; stage-by-stage comparison required at final design.

RESERVOIR ROUTING — EXISTING CROSSING

STORM	INFLOW	OUTFLOW	ATTENUATED	PEAK STAGE
10-year	388 cfs	215 cfs	44.6%	EL 54.31
25-year	606 cfs	512 cfs	15.5%	EL 54.82
50-year	814 cfs	758 cfs	6.8%	EL 55.13
100-year	1,058 cfs	1,032 cfs	2.4%	EL 55.43

CROSSING ALTERNATIVES — DOWNSTREAM TRANSFER

OPTION	10-YR OUT	CHANGE	100-YR OUT	CHANGE
existing 36 in	215	+0	1,032	+0
48 in RCP	179	-36	1,023	-9
72 in RCP	275	+60	981	-52
twin 72 in RCP	333	+118	870	-162
8 x 6 ft box	307	+91	853	-80
10 x 8 ft box	347	+132	869	-163
twin 10 x 8 ft box	388	+173	974	-58
20 x 8 ft box (bridge-class)	388	+173	974	-58
twin 20 x 8 ft box	388	+173	1,058	+26

COMPENSATORY STORAGE SCHEDULE

CELL	LOT	AREA (SF)	MEAN DEPTH	VOLUME
CS-53	53	2,500	1.57 ft	149 cy
CS-54	54	800	0.79 ft	18 cy
CS-55	55	7,400	2.30 ft	630 cy
CS-56	56	800	4.16 ft	123 cy
CS-E	54/55	9,721	10 EL 48.0	153 cy

PROVIDED

REQUIRED by proposed grading

DEFICIT

FILL BELOW EL 55.53 — DISTRIBUTION

beneath dwelling footprints

within 10 ft of a dwelling

driveways, aprons, set-out

RESOLUTION: reduce fill. Vented stem-wall or

pier foundations and driveways at existing grade

bring the balance within the volume provided.

FLOODPLAIN REMOVAL AND FOUNDATION DETERMINATION

LOT	LOWEST GROUND	WS MUST FALL BELOW	REQUIRED DROP	ACHIEVABLE
LOT 53	EL 48.78	EL 48.78	6.69 ft	partial
LOT 54	EL 44.97	EL 44.97	10.50 ft	partial
LOT 55	EL 44.66	EL 44.66	10.81 ft	partial
LOT 56	EL 53.34	EL 53.34	2.13 ft	partial

Controlling talwater lower bound: EL 44.45

The upstream water surface cannot be lowered

below the downstream water surface.

LOWEST FLOOR DETERMINATION

LOT	PROPOSED FF	BASEMENT	REQ LOWEST FLOOR	STATUS
LOT 53	59.24	none	57.52	COMPLIES
LOT 54	57.57	none	57.52	COMPLIES
LOT 55	57.60	none	57.52	COMPLIES
LOT 56	61.02	none	57.52	COMPLIES

Basements are not permissible on any lot.

FREQUENCY OF INUNDATION — EXISTING GROUND

STORM	STAGE	LOT 54	LOT 55	ROAD
2-yr	EL 50.17	89%	41%	clear
5-yr	EL 53.02	97%	79%	clear
10-yr	EL 54.31	99%	92%	OVERTOPPED
25-yr	EL 54.82	100%	93%	OVERTOPPED
50-yr	EL 55.13	100%	95%	OVERTOPPED
100-yr	EL 55.43	100%	95%	OVERTOPPED

RECOMMENDED: DEVELOP LOTS 53 AND 56;

DEDICATE LOTS 54 AND 55 AS FLOODPLAIN

compensation needed by 53 + 56

available on 54 + 55 to EL 50.0

ratio provided to required

31 cy

1,399 cy

45:1 : 1